



CIMA Management Case Study

Mock Exam 4

Cartn

Unseen [3 hours]

Task 1 - Unseen case material [45 minutes]

A new government regulation in Harrland will take effect in 18 months, completely banning the plastic linings used by Cartn, forcing the company into a massive technological shift. To comply, Cartn are currently evaluating a significant investment of H\$15 million in 'Bio-Shield' digital coating machines. This technology replaces the traditional polyethylene plastic layer with a biodegradable, starch-based coating that is applied digitally for precision.

From: Elizabeth Maenda, Senior Financial Manager

To: Financial Manager

Subject: Digital Data Sources and Quality Management

Hi,

The Board is under pressure to improve the recyclability of our cartons due to the new government regulation banning plastic linings. We have already produced a prototype and have received a Quality Control (QC) Report on its performance.

However, the Board is cautious. The initial NPV calculation looks positive, but we need to ensure our data sources for this appraisal are robust and that we have a plan to maintain quality and cost-efficiency if we move to full-scale production.

I've attached an email from the Production Director with the QC report. I need your input on two specific areas for tomorrow's briefing

- Identify three specific digital data sources, internal or external, that Cartn should utilise to improve the accuracy of its capital investment appraisal for the 'Bio-Shield' machines..

[sub-task (a) = 40%]

- Recommend with reasons how Cartn can apply Value Engineering and Total Quality Management (TQM) to the new Bio-Shield product line.

[sub-task (b) = 60%]

Kind regards,
Elizabeth

Reference material

Email from Operations Director

Hi Mario,

Following our recent pilot of the Bio-Shield digital coating technology, my team has completed the formal Quality Control (QC) Report on the prototype units. The digital coating process is a significant departure from our traditional spray-on polyethylene methods. It offers precision, but as the report below indicates, there are technical hurdles we haven't fully cleared yet, particularly concerning our high-volume UHT lines.

Please see the summary of the QC findings below:

INTERNAL QC REPORT: BIO-SHIELD PROTOTYPE (BATCH #402)

Test Category	Performance Metric	Observations & Technical Findings
Material Efficiency	12% Reduction in Waste	Digital application nozzles are far more precise than traditional sprayers, significantly reducing overspray and raw material waste.
Heat Resistance	Marginal Failure at 135°C	During simulated UHT (Ultra-High Temperature) processing, the starch-based bio-coating showed signs of thinning and micro-cracking. This could compromise the 6-month shelf life of UHT milk.
Structural Integrity	4% Leakage Rate	4 out of every 100 units failed the pressure test. This was traced to sensor calibration errors in the digital nozzles, resulting in uneven coating at the carton seams.
Adhesion	Strong Bond on Paperboard	The bio-shield material bonded exceptionally well to the primary paperboard layer, exceeding the performance of traditional plastic laminates in dry conditions.

My Concerns:

My factory managers are already struggling with the volume targets for our standard milk cartons. If we introduce a technology that has a 4% failure rate, we risk a massive spike in internal failure costs and potential recalls. Furthermore, if the thinning at 135°C isn't resolved, the Bio-Shield will be useless for our UHT customers, who make up a massive portion of our recurring manufacturing revenue.

Kind regards, Alison Cruikshank

Task 2 - Unseen case material [45 minutes]

The Board is meeting to discuss the Bio-Shield product launch. While the technical aspects are being refined, the Marketing and PR departments have proposed a radical shift in how Cartn captures value. They believe that simply selling a greener carton is not enough to maintain a competitive advantage; instead, Cartn should lead the industry by adopting a circular 'Take-Back & Recycle' business model.

From: Elizabeth Maenda, Senior Financial Manager

To: Financial Manager

Subject: Circular Business Model and Stakeholders

Hi,

As the recent announcement of this potential shift in business models has leaked to the press, it has caused a mixture of excitement and hostility.

The Marketing Director has attached an executive summary of the 'Take-Back & Recycle' business model and asked for our support in understanding how it will create sustainable value and how to manage key stakeholder groups.

Please can you address the two points below?

- Evaluate the proposed Circular Business Model for the Bio-Shield product range and explain how this model creates sustainable value for Cartn's stakeholders

[sub-task (a) = 60%]

- Identify the key stakeholder groups affected by the transition to the Bio-Shield range and advise the Board on an appropriate communication process for each group.

[sub-task (b) = 40%]

Kind regards,
Elizabeth

Reference material

Executive Summary: Circular Business Model

To: The Board of Directors

From: Haitham Amer, Marketing Director

Subject: Proposal for circular business model

We are evaluating a shift to a circular business model for the Bio-Shield range. Instead of the traditional linear approach of Take-Make-Dispose, we propose a 'Take-Back & Recycle' scheme.

Proposed Mechanics:

- **Partnership:** Partner with major supermarket retailers to install Cartn-Collection Points.
- **Repurposing:** Cartn will collect used Bio-Shield cartons, transport them to a dedicated facility to separate the layers, and re-pulp the paper fibres.
- **Secondary value:** These fibres will be used to manufacture non-food secondary packaging (e.g., corrugated cardboard for shipping) or sold to building material manufacturers.
- **Pricing shift:** Transition from unit-sales to a 'Packaging-as-a-Service' subscription, covering the supply of cartons and the cost of the end-of-life take-back logistics.

Public Relations Concern:

Rumours are circulating in the Harrland Daily News about our potential 'Take-Back & Recycle' scheme. While some coverage is positive, the following challenges have emerged:

1. **Environmental Groups (Green-Harrland):** They have issued a press release calling our scheme a marketing gimmick. They argue that unless we can recycle the plastic/aluminium layers back into food-grade packaging, we are merely downcycling.
2. **Small Dairy Farmers Consortium:** We received an urgent letter from a group of small-scale milk producers. They are fearful that the high cost of the Bio-Shield subscription - and the logistical burden of the take-back scheme - will drive them out of business, favouring only the big dairy conglomerates who have the margins to absorb these costs.

Task 3 - Unseen case material [45 minutes]

Following a 3-month pilot of the Bio-Shield project in Central City, the project committee has raised concerns over the return rate and the viability of the project being a success.

From: Elizabeth Maenda, Senior Financial Manager

To: Financial Manager

Subject: Measuring Risk and Project Management Tools and Techniques

Hi,

We are currently facing the perfect storm of operational and financial hurdles as the Bio-Shield project is ramping up.

The Finance Director has forwarded me an email from the Project Committee and is concerned with the findings from the 3-month pilot.

Please can you prepare a report on the following risk management measures?

- Recommend and explain how Cartn can apply Sensitivity Analysis and Scenario Planning to evaluate the viability of the new Bio-Shield circular business model

[sub-task (a) = 40%]

- Evaluate and recommend three project management tools or techniques that should be implemented to manage the project effectively

[sub-task (b) = 60%]

Kind regards,
Elizabeth

Reference material

Project Committee Email

To: Mario Baquero, Finance Director

From: Project Lead, Bio-Shield Circular Initiative

Date: 1 April 2026

Subject: URGENT: Bio-Shield Infrastructure Failure & Regulatory Risk

Mario,

We have completed the 3-month "Take-Back" pilot in Central City, and the results indicate that the current project trajectory is unsustainable due to the risks highlighted below.

1. Pilot Performance & Viability Gap

The consumer return rate during the pilot was only 22%, significantly below the 40% threshold required for the logistics to be cost-neutral. Furthermore, a confidential legal briefing suggests that the upcoming "Harrland Green-Tax" will only exempt starch-based products if we can prove a 50% recovery rate. If we fail to hit this 50% target, we face a H\$500 per tonne tax, effectively neutralising our competitive price advantage.

2. IT & Logistics Deadlock

The IT tracking system is now 3 months behind schedule. Friction between IT (focusing on high-security encryption) and Logistics (claiming the interface is too complex for drivers) has stalled development. We have no formal mechanism to resolve these cross-functional disputes.

3. Scope Creep & Budget Overrun

The project is suffering from significant scope creep due to unauthorised feature requests from the marketing department - most notably a late-stage requirement for individual customer carbon-tracking. This has caused the IT budget to balloon from H\$1.2 million to H\$2.8 million. Currently, these changes are being requested via email directly to developers, bypassing both the Project Manager and Finance.

Task 4 - Unseen case material [45 minutes]

As Cartn prepares for the full-scale rollout of the Bio-Shield range, the Finance and Operations departments have identified significant legacy issues at the oldest manufacturing plant, Plant Alpha.

From: Elizabeth Maenda, Senior Financial Manager

To: Financial Manager

Subject: Plant Alpha Accounting Treatment and Conflict Resolution

Hi,

The Finance Director has been alerted to the issues at one of our oldest manufacturing plants and would like clarity on the accounting treatment we should apply to the soon-to-be obsolete extruders.

Furthermore, he is concerned that the internal conflict between sales and manufacturing will stall the roll-out of the Bio-Shield project.

Please can you support the Finance Director with a report on the following;

- Recommend the appropriate accounting treatment for the extruders and the site clean-up commitment, and explain the impact these treatments will have on Cartn's financial statements

[sub-task (a) = 60%]

- Recommend two specific conflict resolution strategies that the Board should implement to resolve this conflict and ensure both departments work together toward the plastic-free transition.

[sub-task (b) = 40%]

Kind regards,
Elizabeth

Reference material

Email from Finance Director

To: Elizabeth Maenda, Senior Financial Manager

From: Mario Baquero, Finance Director

Subject: Plant Alpha transition: Accounting and Personnel Issues

We have been made aware of urgent matters regarding the conversion of Plant Alpha to a Bio-Shield facility:

Technical obsolescence of extrusion assets

Our traditional lamination extruders at Plant Alpha currently have a carrying amount of H\$8.4 million. These were originally expected to operate until 2032. However, following the recent Board decision to cease all plastic-laminated production by the end of next year, these assets will be decommissioned in 18 months. We have also received a quote from a scrap metal firm offering only H\$200,000 for the units in 18 months, which is significantly below our projected residual value.

The Clean-Harmland public commitment

Last week, our CEO, Guiquan Li, gave a televised interview where he formally committed Cartn to a site decontamination program for Plant Alpha. The local community is concerned about chemical residues from 40 years of plastic processing. Our environmental engineers estimate the cleanup will cost H\$3.2 million, likely payable in two years. Although no legislation currently forces us to clean this specific site, the Board feels the public announcement has created an expectation we cannot ignore.

Breakdown in functional collaboration

Haitham Amer (Marketing/Sales) recently walked out of a meeting, claiming that Alison Cruikshank (Production) is deliberately sabotaging his sales targets by padding the Bio-Shield production budget with unnecessary green overheads.

Haitham argues that his team cannot sell a product that is 25% more expensive than our competitors' traditional cartons. Alison has retaliated by accusing the sales team of strategic negligence, stating that if we don't transition now, the upcoming ban by the Harmland government on plastic linings will lead to the total closure of the company. This impasse is preventing us from signing several long-term dairy contracts.

Present value tables

Present value of US\$1 i.e. $(1 + r)^{-n}$ where r = interest rate, n = number of periods until payment or receipt.

Period s (n)	Interest rates (r)									
	1%	2%	3%	4%	5%	6%	7%	8%	9%	10%
1	.990	.980	.971	.962	.962	.943	.935	.926	.917	.909
2	.980	.961	.943	.925	.907	.890	.873	.857	.842	.826
3	.971	.942	.915	.889	.864	.840	.816	.794	.772	.751
4	.961	.924	.888	.855	.823	.792	.763	.735	.708	.683
5	.951	.906	.863	.822	.784	.747	.713	.681	.650	.621
6	.942	.888	.837	.790	.746	.705	.666	.630	.596	.564
7	.933	.871	.813	.760	.711	.665	.623	.583	.547	.513
8	.923	.853	.789	.731	.677	.627	.582	.540	.502	.467
9	.914	.837	.766	.703	.645	.592	.544	.500	.460	.424
10	.905	.820	.744	.676	.614	.558	.508	.463	.422	.386
11	.896	.804	.722	.650	.585	.527	.475	.429	.388	.350
12	.887	.788	.701	.625	.557	.497	.444	.397	.356	.319
13	.879	.773	.681	.601	.530	.469	.415	.368	.326	.290
14	.870	.758	.661	.577	.505	.442	.388	.340	.299	.263
15	.861	.743	.642	.555	.481	.417	.362	.315	.275	.239
16	.853	.728	.623	.534	.458	.394	.339	.292	.252	.218
17	.844	.714	.605	.513	.436	.371	.317	.270	.231	.198
18	.836	.700	.587	.494	.416	.350	.296	.250	.212	.180
19	.828	.686	.570	.475	.396	.331	.277	.232	.194	.164
20	.820	.673	.554	.456	.377	.312	.258	.215	.178	.149

Period s (n)	Interest rates (r)									
	11%	12%	13%	14%	15%	16%	17%	18%	19%	20%
1	.901	.893	.885	.877	.870	.862	.855	.847	.840	.833
2	.812	.797	.783	.769	.756	.743	.731	.718	.706	.694
3	.731	.712	.693	.675	.658	.641	.624	.609	.593	.579
4	.659	.636	.613	.592	.572	.552	.534	.516	.499	.482
5	.593	.567	.543	.519	.497	.476	.456	.437	.419	.402
6	.535	.507	.480	.456	.432	.410	.390	.370	.352	.335
7	.482	.452	.425	.400	.376	.354	.333	.314	.296	.279
8	.434	.404	.376	.351	.327	.305	.285	.266	.249	.233
9	.391	.361	.333	.308	.284	.263	.243	.225	.209	.194
10	.352	.322	.295	.270	.247	.227	.208	.191	.176	.162
11	.317	.287	.261	.237	.215	.195	.178	.162	.148	.135
12	.286	.257	.231	.208	.187	.168	.152	.137	.124	.112
13	.258	.229	.204	.182	.163	.145	.130	.116	.104	.093
14	.232	.205	.181	.160	.141	.125	.111	.099	.088	.078
15	.209	.183	.160	.140	.123	.108	.095	.084	.074	.065
16	.188	.163	.141	.123	.107	.093	.081	.071	.062	.054
17	.170	.146	.125	.108	.093	.080	.069	.060	.052	.045
18	.153	.130	.111	.095	.081	.069	.059	.051	.044	.038
19	.138	.116	.098	.083	.070	.060	.051	.043	.037	.031
20	.124	.104	.087	.073	.061	.051	.043	.037	.031	.026

Cumulative present value tables

This table shows the Present Value of US\$1 per annum, Receivable or Payable at the end of each year for n years

$$\frac{1 - (1+r)^{-n}}{r}$$

Periods (n)	Interest rates (r)									
	1%	2%	3%	4%	5%	6%	7%	8%	9%	10%
1	0.990	0.980	0.971	0.962	0.952	0.943	0.935	0.926	0.917	0.909
2	1.970	1.942	1.913	1.886	1.859	1.833	1.808	1.783	1.759	1.736
3	2.941	2.884	2.829	2.775	2.723	2.673	2.624	2.577	2.531	2.487
4	3.902	3.808	3.717	3.630	3.546	3.465	3.387	3.312	3.240	3.170
5	4.853	4.713	4.580	4.452	4.329	4.212	4.100	3.993	3.890	3.791
6	5.795	5.601	5.417	5.242	5.076	4.917	4.767	4.623	4.486	4.355
7	6.728	6.472	6.230	6.002	5.786	5.582	5.389	5.206	5.033	4.868
8	7.652	7.325	7.020	6.733	6.463	6.210	5.971	5.747	5.535	5.335
9	8.566	8.162	7.786	7.435	7.108	6.802	6.515	6.247	5.995	5.759
10	9.471	8.983	8.530	8.111	7.722	7.360	7.024	6.710	6.418	6.145
11	10.368	9.787	9.253	8.760	8.306	7.887	7.499	7.139	6.805	6.495
12	11.255	10.575	9.954	9.385	8.863	8.384	7.943	7.536	7.161	6.814
13	12.134	11.348	10.635	9.986	9.394	8.853	8.358	7.904	7.487	7.103
14	13.004	12.106	11.296	10.563	9.899	9.295	8.745	8.244	7.786	7.367
15	13.865	12.849	11.938	11.118	10.380	9.712	9.108	8.559	8.061	7.606
16	14.718	13.578	12.561	11.652	10.838	10.106	9.447	8.851	8.313	7.824
17	15.562	14.292	13.166	12.166	11.274	10.477	9.763	9.122	8.544	8.022
18	16.398	14.992	13.754	12.659	11.690	10.828	10.059	9.372	8.756	8.201
19	17.226	15.679	14.324	13.134	12.085	11.158	10.336	9.604	8.950	8.365
20	18.046	16.351	14.878	13.590	12.462	11.470	10.594	9.818	9.129	8.514
Periods (n)	Interest rates (r)									
	11%	12%	13%	14%	15%	16%	17%	18%	19%	20%
1	0.901	0.893	0.885	0.877	0.870	0.862	0.855	0.847	0.840	0.833
2	1.713	1.690	1.668	1.647	1.626	1.605	1.585	1.566	1.547	1.528
3	2.444	2.402	2.361	2.322	2.283	2.246	2.210	2.174	2.140	2.106
4	3.102	3.037	2.974	2.914	2.855	2.798	2.743	2.690	2.639	2.589
5	3.696	3.605	3.517	3.433	3.352	3.274	3.199	3.127	3.058	2.991
6	4.231	4.111	3.998	3.889	3.784	3.685	3.589	3.498	3.410	3.326
7	4.712	4.564	4.423	4.288	4.160	4.039	3.922	3.812	3.706	3.605
8	5.146	4.968	4.799	4.639	4.487	4.344	4.207	4.078	3.954	3.837
9	5.537	5.328	5.132	4.946	4.772	4.607	4.451	4.303	4.163	4.031
10	5.889	5.650	5.426	5.216	5.019	4.833	4.659	4.494	4.339	4.192
11	6.207	5.938	5.687	5.453	5.234	5.029	4.836	4.656	4.486	4.327
12	6.492	6.194	5.918	5.660	5.421	5.197	4.968	4.793	4.611	4.439
13	6.750	6.424	6.122	5.842	5.583	5.342	5.118	4.910	4.715	4.533
14	6.982	6.628	6.302	6.002	5.724	5.468	5.229	5.008	4.802	4.611
15	7.191	6.811	6.462	6.142	5.847	5.575	5.324	5.092	4.876	4.675
16	7.379	6.974	6.604	6.265	5.954	5.668	5.405	5.162	4.938	4.730
17	7.549	7.120	6.729	6.373	6.047	5.749	5.475	5.222	4.990	4.775
18	7.702	7.250	6.840	6.467	6.128	5.818	5.534	5.273	5.033	4.812
19	7.839	7.366	6.938	6.550	6.198	5.877	5.584	5.316	5.070	4.843
20	7.963	7.469	7.025	6.623	6.259	5.929	5.628	5.353	5.101	4.870